

Jake Fogel

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EDUCATION

New York University, Tandon School of Engineering , New York, NY Master of Science in Computer Science	<i>Expected May 2028</i>
McGill University , Montreal, QC Bachelor of Science in Computer Science <i>Relevant Coursework: Algorithms & Data Structures, Applied Machine Learning, Computational Linguistics, Software Design</i>	<i>Sep 2022 – May 2026</i>
Sophia University , Tokyo, Japan Exchange Program (Spring Semester)	<i>Mar 2025 – Aug 2025</i>

EXPERIENCE

Global SAP Innovation Intern – Syntax	<i>May 2024 – Aug 2024</i>
- Developed a Software as a Service (SaaS) platform using LLM agents and advanced prompting techniques (self-consistency, chain of thought) to automate the creation of planning documents, code, and test cases for SAP RICEFs. - Engineered a high-efficiency Python API with FastAPI, integrating a custom Retrieval-Augmented Generation (RAG) system via ChromaDB to manage state persistence and document parsing. - Deployed the Python API on an SAP Cloud Foundry server, streamlining CI/CD pipelines for automated testing. - Authored comprehensive technical documentation, reducing future developer onboarding time.	
Junior Software Developer – Advantage Group International	<i>May 2023 – Aug 2023</i>
- Developed JavaScript workflow rules, following Agile methodology, to automate the company-wide ticketing system (YouTrack) increasing transaction efficiency by 152% on average. - Designed mock-ups for Advantage survey report slides, providing valuable input for the final Tableau implementation to enhance data visualization and presentation quality. - Served on a 3-person selection committee evaluating candidates for the Junior Software Developer role.	
Director, Game Design Camp – UCC Summer Camps	<i>Jun 2022 – Aug 2022</i>
- Hired and led a team of 5 staff, giving regular performance evaluations. - Created and delivered comprehensive C# and Unity lessons, increasing program efficiency via improved training material.	

PROJECTS

Odyssey Goal Tracking Application (Flutter, Dart, Firebase)	<i>March 2026 – Jun 2026</i>
- Engineered a scalable mobile application using Flutter and Dart, integrating Firebase Authentication and Cloud Firestore to ensure secure, real-time data synchronization. - Designed interactive user interfaces incorporating glassmorphism and dynamic progress timelines, implementing custom state management algorithms to track daily streaks and milestone progression.	
Contextual Emotion Recognition Engine (Python, Hugging Face, BERT)	<i>Sep 2024 – Dec 2024</i>
- Engineered a robust NLP text classification pipeline to process the imbalanced GoEmotions dataset of Reddit comments. - Fine-tuned state-of-the-art transformer models (RoBERTa, DistilBERT) to extract semantic representations, achieving a 62% classification accuracy (73% improvement over baseline models). - Leveraged Word2Vec embeddings and tree-based algorithms (XGBoost, Random Forest) for baseline benchmarking.	
Medical Image Classifier (Python, PyTorch)	<i>Sep 2024 – Dec 2024</i>
- Built a custom convolutional neural network to identify organ tissue from CT scan data, achieving 89.35% test accuracy (27.3% improvement over baseline perceptron). - Applied transfer learning by fine-tuning a pre-trained ResNet18 model to achieve 84.85% test accuracy, conducting hyperparameter searches on learning rates and regularization.	
Simulated Operating System (C, Linux/UNIX)	<i>Sep 2025 – Dec 2025</i>
Developed a multi-component operating system in C, including a custom shell, multi-process scheduler (FCFS, SJF, RR, AGING), and a demand paging memory manager with an LRU replacement policy.	

ACTIVITIES & LEADERSHIP

Experience Team Lead , McHacks Hackathon (McGill University)	<i>Sep 2024 – Present</i>
- Manage McGill's largest (700+ participant) hackathon, leading recruitment, workshops, seminars, and panels. - Negotiate partnerships with hotel and transportation vendors to secure sponsorships and benefits for participants.	

TECHNICAL SKILLS

Languages: Python, Java, C++, C, JavaScript, SQL, Dart, HTML/CSS
Frameworks & Libraries: Flutter, FastAPI, React, Node.js, PyTorch, Hugging Face, Scikit-learn, XGBoost, NumPy
Tools & Platforms: Git, Bash, Linux/UNIX, Docker, Firebase/Firestore, ChromaDB, GitHub, Unity